sheet is carrying equally the same amount of pressure per square inch as the sheet in the fire-box, still the ends in the front flue sheet are simply rolled. We might, therefore, reasonably assume that the bead is of but little value in supporting and protecting the sheet against sagging and bulging.

This brings us to the consideration of the endurance of the bead. While it is true that the bead is gradually wasting away by reason of the inpingement of the heat from the fire-box gases, it is also true that not only are particular forms of tools responsible for certain developments, but quite an equal amount can be attributed to the manner of their manipulation. We refer to the bead wasting away because its vitality is gradually decreasing. It may possibly appear under certain conditions that the bead is growing larger, but the stock is being drawn from the body of the flue. It might be well to here touch upon the difference between the effect of the roller and the prosser, the difference in the effect of the prosser with and without the boss, also the effect of the beading tool. A number of experiments seem to indicate that while there is possibly a little difference in the ultimate life of a flue when rolled as compared with one prossered, it is found that the roller has the decided effect of drawing the bead from the sheet, making it necessary that the stock in the bead be again closed down, by hand or otherwise, to again lay it up to the sheet. The tendency of the flue to draw is not entirely absent under the prosser, but less pronounced. It is also a fact, we believe, that a prosser, at least those made with the boss, can only be used for a certain period during the life of the flue. This is not true of the roller nor of the prosser where the boss has been left off. Where the bead is decreasing in size, judgment must be exercised in the use of the beading tool to prevent guttering of the flue sheet. It also requires judgment to decide just how long this latter tool can be used before it has to be given up entirely for the roller. This, of course, obtains during the latter portion of the life of the flue. It is quite possible about this period in the life of a flue to sacrifice flue sheets for flue beads on account of the manner the flues are worked and maintained, and this critical point would seem to be after about one-half the vitality of the bead had been consumed. At this period the bead has become weakened by excessive working, and perhaps has shown indications of cracking and otherwise deteriorating. As a result of a number of carefully concluded experiments, it has not been definitely shown that in the end the roller is detrimental, and therefore expensive. Referring to the sectional prosser with and without the boss, probably twenty years ago it was the idea that the boss was essential, and that as one of its functions it helped to stay the flue sheet, together with perfecting the joint on the water side, but if this was a necessity with boilers carrying 120 to 140 pounds pressure, the theory falls flat when it is asserted, and even proven conclusively, that the prosser with the boss left off is being used exclusively in boilers carrying 200 pounds pressure and upward.

Having some doubt as to the real value of the above theories, your committee has made several laboratory experiments with a view of ascertaining the holding power of different types of settings of flues, with and without beads, as well as with and without ferrules, rolled and prossered, also to determine practically the possibility of using unbeaded flues in locomotive service. Such tests have been carried along uniformly for the past fifteen months on several 100-ton consolidation freight locomotives carrying a working pressure of 200 pounds. The particular engines in question are equipped with 258 2¼-inch flues 16 feet 6 inches long. The flues were placed flush with the fire-box side of the flue sheet, making no attempt whatever to bead them. These experiments were simplified by making them on locomotives equipped with the so-called re-enforcing sheet. Nothing is claimed for the re-enforcing flue sheet in these particular experiments in the way of increasing the endurance of the bead, but by construction it permitted the flues to be so carried as to prevent any possibility of the flues pulling through the fire-box flue sheet. In among these 258 flues, fifteen flues were put in with beads, locating them regularly throughout the lower central portion of the set. While it can be said that the reports show that all of the flues required a certain amount of attention, it is also true that the number of times the unbeaded flues were worked upon, as compared with the beaded flues, is as 1 to 2.3. Encouraged by these experiments, and being able to continue in service engines with the beads left off, other engines have been fitted up in like manner, and the outcome has been that the fifteen beaded flues in the experiment above referred to gradually broke down by deterioration, but not until they had given some eight months' service; on the other hand, the unbeaded flues are still holding very satisfactorily, and have been in service some fourteen months. Upon close investigation it was found that the unbeaded flues were about normal. From the above we might conclude that the bead has no particular value towards preventing the flue from burning off; on the contrary, the beads have apparently picked up heat, and possibly in proportion to the size of the gradually increasing stock in the bead, aggravated by the thickness of the flue sheet, as well as any deposit of scale and mud, and have burned off, as might be expected. It might also be interesting to mention that while the unbeaded flues were flush with the flue sheet, they have, by reason of the rolling process, been drawn to possibly as much as 1/16 inch beyond the fire side of the flue sheet.

Probably in many localities throughout the country the bead determines the life of the flue, except where the water changes the conditions. It is generally known that working iron or steel, as the case may be, will harden and toughen the material, destroying its ductility, and hence under the present method of working the bead, however it is done, is from its initial application to its removal undergoing constant and severe punishment. The constant vibration of the flue undoubtedly sets up a movement in the flue sheet, resulting in not only drawing the bead and otherwise disturbing the original joint, but must also force the flue inwardly, and this theory is ventured because it is not uncommon to find the copper ferrules thinned out to twice their original length and driven inwardly. As this movement continues, the bead is gradually drawn from the sheet, increasing its exposure and decreasing its resistance to the impinging flames. In order to overcome this, the bead must necessarily be subjected to constant working and re-beading. This work produces incipient cracks in the bead, which occur as the material is gradually weakened by crystallization, and we might reasonably infer that as the material is alternately opened and closed, a certain amount of impurities find their way into the bead, all of which has been pretty well determined, resulting in breaking them down and otherwise destroying them after obtaining remarkably low mileage. The bead in its present condition is certainly a victim of circumstances, and mainly an account of the little protection it receives from the water. There is no disputing the fact that the circulation of water, however imperfect, affords some protection, but it is also true that a poor conduction of heat adds an incalculable amount towards the deterioration of the flue bead, together with the many disadvantages by reason of treatment and limitations in construction.

An act is now before the Legislature of the State of Wyoming to establish a State Board for the inspection of steam boilers and for the examination and licensing of engineers. The bill provides for an annual inspection of boilers and a hydrostatic test in excess of the working pressure.

so doing will still remain as right. We do not think, however, that the subject should be passed by without going into the matter more thoroughly. Practically all of the devices pertaining to the locomotive have been improved, and we cannot say without testing the matter but what a different location of the check and the water being deposited above the waterline would be better than below the waterline. Most likely this test could be brought about by our representatives to the Master Mechanics' Convention, who, in our opinion, are the proper people to have this test made.

As to the steaming of engine and working of injector, your committee fully agrees with all the mechanical advice it has that there is no noticeable difference in any location that should be made; however, the checks and injectors should be placed so as to allow slow, easy bends in the branch pipe.

The following is from the personal experience of your chairman: With the operation of the checks above the waterline, the committee would recommend that all checks be placed as far from the throttle-box as possible, due to the fact that with a check above the waterline close to the throttle-box, the throttle siphons the water very noticeably when the injector is working, which is bad on the valves and packing. We have, however, handled engines with the checks located on top of boiler and on front sheet next to smoke-box and noticed no bad results whatever along this line when the injectors were working. We would also suggest that no iron pipes be used with such location of checks, due to the fact that the bend that is necessary to be made to the check on top of boiler makes it next to impossible to keep the joints tight, due to expansion.

The following remarks are concerning the application and practices that should be followed in applying feed pipes, check valves, etc.:

The free and unrestricted passage of the water from the tank to the injector of a locomotive is of the utmost importance, and to attain this object it is necessary that all auxiliaries in connection with the water-feeding apparatus should be of proper construction, location and size.

One of the most important parts of the water-feeding system is the boiler check valve, and, in connection with it, the feed-water delivery pipe.

The pipe should be as short as circumstances and a well-established practice allow, and as straight as possible. In locomotive practice, the feed-water delivery pipes are usually bent, but these bends should not contain sharp elbows, but should be made in easy curves, so as to avoid abrupt changes in the current of the water and resulting frictional resistance, which may affect the general performance of the injector. For the same reason the pipe should be of proper size and free of obstructions of any kind, as obstructions in the delivery pipes will create a back pressure against the injector, which may not only affect its general efficiency, but may cause the water to be thrown out through the overflow, resulting in considerable waste of water if the injector is provided with an open overflow; or, in case of a closed overflow, injector obstructions may cause the injector to fly off, with the result that live steam will blow back into the tank, eventually heating the water to an extent that will make it difficult to again start the injector until opportunity is found to cool off the water.

As to the size of the delivery pipe, it is safe practice to follow the recommendations of the manufacturers of injectors and to make the pipe of the size called for by the connections at the delivery end of the injector.

Concerning the boiler check the first requisite is that it should be of ample size. In their connection the size of the pipe connections provided for the injectors by their makers give a safe connection.

All passages in the boiler check should be of the full size called for by these connections.

The check valve should have a sufficient lift as to avoid crowding and stowing of the water, which should result in back pressure and may be followed by consequences as related above. A lift of one-quarter to one-third of the clear opening covered by the check valve will be found serviceable, provided the clear opening is of the proper size. Preferably the boiler check should be of the "straight" type; that is, the water from the delivery pipe should enter the check body underneath the check valve, passing through it in a straight line, so as to avoid "turns" in the direction of the flow of water as much as possible.

The boiler check should be provided with a removable seat, as thereby the body of the valve is made practically indestructible, and in case the seat gets worn out, only the seat is to be replaced. The check valve itself and its seat should be accessible for inspection and regrinding or reseating—with pressure in the boiler. For this purpose, the boiler check should be provided with a stop valve; that is, a valve which, when shut, cuts off communication between the boiler and the check chamber, permitting the opening of the check chamber by removing the caps usually provided for this purpose. In case of double boiler checks, the two check chambers should be independent and separate from each other, and each chamber should be provided with its own stop valve, so that in case one side should become inoperative from any reason it could be cut off without interfering with the proper operation of the other side.

The stop valve should preferably be placed in a valve chamber of its own, independent of the check chamber, so that the stop valve could not possibly interfere with the proper operation of the check valve or prevent access to the check valve. If the stop valve is located in a chamber common to it and the check valve, the latter is not accessible, except after removal of the stop valve or after disconnecting the pipe attached to the boiler check, either of which necessities nullify the true purpose and usefulness of a stop valve.

THE MODERN LOCOMOTIVE BOILER.*

BY JOHN W. HOBSON.

JOINTS AND RIVETING.

The joints in a locomotive boiler should receive great consideration, because they are subjected to severe stresses as well as to a continual vibration. In this country, as a rule, the longitudinal joints, throat plate joints and smoke-box barrel ring are double riveted, the remainder being single riveted. The longitudinal joints should be well out of the water, to avoid corrosion. Of course, in the case of small boilers this is sometimes impossible on one ring; then it is better to have the joint totally submerged. The longitudinal joints should always be butt-jointed with double straps, as, if they are lap-jointed, grooving is almost sure to set in along the edge of the inner lap. This grooving is caused partly by the mechanical motion of the joint as the plate tends to pull into a straight line and partly by the chemical action of the feed water. The longitudinal lap-joint has led to more explosions than any other cause.

There is still a good deal of difference of opinion as to whether the rivet holes should be punched or drilled. It is now more common for them to be drilled, and there is very little difference in cost, because a punched plate must be annealed after punching, and it requires more handling to get

* Continued from the October issue.